

ALGEBRA QUALIFYING EXAMINATION

Fall 2024

PLEASE WRITE YOUR UO ID NUMBER ON YOUR SOLUTIONS but not your name.

There are three parts to this exam:

- (20 points) Part I is asking for statements of some definitions/theorems. Two questions, give concise but accurate answers.
- (64 points) Part II is True/False. Four questions, give a sketch proof or counterexample for full credit.
- (20+96 points) Part III consists of one medium-length problem worth 20 points then three longer problems worth a total of 96 points. Give concise but clear proofs stating clearly any results you are using.

ATTEMPT ALL QUESTIONS

Part I. Briefly state each of the following:

1. For a functor $F : \mathcal{C} \rightarrow \mathcal{D}$, what it means for F to be an equivalence of categories.
2. For an R -module M , two different but equivalent definitions of what it means for M to be injective.

Part II. True or false? Provide a brief sketch of a proof (of truth or falsity), or provide a counterexample where relevant:

1. If U, V are finite-dimensional vector spaces and $f : U \rightarrow V, g : V \rightarrow U$ are linear maps then $\text{trace}(f \circ g) = \text{trace}(g \circ f)$.
2. $\mathbb{Q}$ is a projective $\mathbb{Z}$ -module.
3. Let M be a finitely generated module over a PID R . If $M/IM = \{0\}$ for all maximal ideals I of R then $M = \{0\}$.
4. There is a ring isomorphism $\mathbb{C}[x, y]/(x^2 - y^2) \cong \mathbb{C}[x, y]/(x - y) \times \mathbb{C}[x, y]/(x + y)$.

Part III. Give complete answers to the following. In the longer questions 2–4, parts marked with a (*) indicate that they rely on the successful completion of an earlier part of the problem; parts without a (*) can be accomplished even without succeeding on the previous parts, though they may rely on assertions made in earlier parts.

1. Let R be a ring. Prove that R is left semisimple if and only if every left ideal of R is generated by an idempotent.
2. This problem is about groups of size 21.
 - (i). Prove that any group G of size 21 is isomorphic to a semidirect product $C_3 \rtimes_{\phi} C_7$.
 - (ii). Classify all groups of size 21 up to isomorphism.
 - (iii). For the rest of this problem, choose a non-abelian group G of size 21. Write down a presentation of G .
 - (iv). Compute the sizes of all conjugacy classes in G . Hint: this can be done by hand, but you can save a lot of time with numerology instead! Note that G has a normal subgroup.
 - (v). Compute the size of the abelianization $G/[G, G]$. Hint: again, you can do this by hand, but you do not need to!
 - (vi). (*) Compute the dimensions of all the irreducible representations of G .
3. We provide an incomplete character table for a mysterious group G .

C	id	C_1	C_2	C_3	C_4	C_5	C_6
order	1	1	4	2	4	2	2
$\mathbb{1}$	1	1	1	1	1	1	1
L_1	1	1	1	1	-1	-1	-1
L_2	1	1	-1	1	1	-1	-1
L_3	1	1	-1	1	-1	1	1
X	2	2	0	-2	0	0	0
Y_1	2	-2	0	0	0	α	β
Y_2	2	-2	0	0	0	γ	δ

- (i). Compute the kernel of the representation X (i.e. the subgroup of elements which act on X trivially).
- (ii). Is $\{id\} \cup C_1 \cup C_3 \cup C_4$ a normal subgroup? Justify your answer.
- (iii). Is $\{id\} \cup C_1 \cup C_3$ a normal subgroup? Justify your answer.

- (iv). Decompose $X \otimes X$ into irreducible representations. Hint: You can guess the answer without computing many pairings, if you look at the column of C_3 .
- (v). Which irreducible representation is L_2^* ?
- (vi). I claim that $Y_1^* \cong Y_2$. Compute the values of α , β , γ , and δ . Note that you can not distinguish between Y_1 and Y_2 with the information given, so you can choose which is which.

4. Let $\alpha : \mathbb{C}[s, t, u] \rightarrow \mathbb{C}[x, y, z]$ be the ring homomorphism $s \mapsto xy$, $t \mapsto xz$, $u \mapsto yz$.

- (i). Let ϕ be the corresponding morphism between algebraic sets. Describe ϕ explicitly.
- (ii). (*) Note that $\phi(1, 2, 3) = (2, 3, 6)$. Describe the preimage $\phi^{-1}(2, 3, 6)$. Hint: if you know the values of xy and xyz then you know the value of z .
- (iii). (*) Describe the preimage $\phi^{-1}(0, 1, 2)$.
- (iv). (*) Consider the preimage $\phi^{-1}(0, 0, 0)$. Describe the decomposition of this preimage into irreducible components, and indicate the prime ideal corresponding to each component.
- (v). The map α is injective, so it embeds $\mathbb{C}[s, t, u]$ into $\mathbb{C}[x, y, z]$. Prove that this is *not* an integral extension.
- (vi). Prove that there is a $\mathbb{C}$ -algebra homomorphism $\beta : \mathbb{C}[s, s^{-1}, t, t^{-1}, u, u^{-1}] \rightarrow \mathbb{C}[x, x^{-1}, y, y^{-1}, z, z^{-1}]$ induced by α .
- (vii). The map β is injective, so it embeds

$$\mathbb{C}[s, s^{-1}, t, t^{-1}, u, u^{-1}] \quad \text{into} \quad \mathbb{C}[x, x^{-1}, y, y^{-1}, z, z^{-1}].$$

Prove that this is an integral extension.